

ATPL EASA STUDY GUIDE | 040 - HUMAN PERFORMANCE

Chapter 8: Information Processing & Memory

Syllabus Source: CAE Oxford (pp. 237-278) | Pass Target: >= 90% AviationExam

1. THE HUMAN INFORMATION PROCESSING MODEL

Human cognition in the cockpit operates as an open-loop information system (Wickens' Model): Sensory Inputs (Vision, Hearing, Proprioception) -> Sensory Memory Store -> Selective Attention & Perception -> Working Memory & Decision Making -> Motor Action (Effector Organs) -> Environmental Feedback.

- Sensory Memory Store: Ultra-short duration: Iconic (visual) store lasts ~0.5 seconds; Echoic (auditory) store lasts ~2 to 4 seconds. Holds raw, unfiltered physical stimuli.
- Perception: The process of interpreting sensory signals by comparing them with past mental models stored in long-term memory. Vulnerable to cognitive expectations and confirmation bias.
- Cognitive Bottleneck: Human conscious processing is SINGLE-CHANNEL and capacity-limited. Simultaneous conscious tasks compete for central cognitive resources, causing channel overload.

2. HUMAN MEMORY ARCHITECTURE (WORKING VS LONG-TERM)

Memory System	Capacity, Duration & Storage Characteristics	Operational Application & Cockpit Vulnerability
Working Memory (Short-Term Memory)	Capacity: 7 ± 2 items (Miller's Law). Duration: 10 to 20 seconds without rehearsal.	Highly vulnerable to disruption and cognitive distraction. Rehearsal and chunking (e.g. transponder code '7-5-0-0' as one chunk) expand functional capacity.
Long-Term Memory: Semantic	Facts, concepts, rules, language, airspace classes.	Deeply encoded, organized into mental schema / neural networks. Resilient to decay.
Long-Term Memory: Episodic	Personal experiences, specific past flight events.	Context-dependent memory of past flights, near misses, and simulator training.
Long-Term Memory: Procedural	Motor flight skills, crosswind landings, scans.	Automated 'motor programs'. Executed with minimal conscious attention. Can revert under acute stress!

[!] EXAM TRAP (AVIATIONEXAM) - Miller's Magical Number 7 ± 2 and Cockpit Chunking

- Working Memory Capacity: George Miller demonstrated that human working memory can hold between 5 and 9 items (7 ± 2) simultaneously.
- Chunking: Grouping individual pieces of data into meaningful units. For example, the instruction 'Maintain FL 350, heading 240, speed 280 kt, squawk 4215' contains 13 individual digits. Unchunked, it exceeds working memory capacity. Experienced pilots chunk it into 4 operational concepts, enabling accurate retention!

3. ATTENTION MECHANISMS & COGNITIVE TUNNELING

Attention is the active cognitive mechanism that directs conscious mental processing to selected stimuli:

- **Selective Attention:** Directing focus to one source of information while filtering out irrelevant background noise (e.g. listening to own callsign on busy ATC frequency - 'cocktail party effect').
- **Divided Attention (Time-Sharing):** Rapidly switching attention across multiple operational tasks (e.g. instrument cross-check scan: Attitude -> Speed -> Altitude -> Heading). True simultaneous conscious processing is an illusion.
- **Cognitive Tunneling (Channelized Attention):** Under high stress, anxiety, or high workload, attention narrows onto one single indicator or problem, completely blinding the pilot to other critical alarms (e.g. Eastern Airlines 401 crash while fixing a landing gear light bulb).

4. MOTOR PROGRAMS & NEGATIVE HABIT TRANSFER

With practice, flight actions become automated motor programs. This frees working memory for strategic decision-making. However, when transitioning between different aircraft types (e.g. from conventional control yoke to sidestick, or clockwise to counterclockwise switches), pilots under stress frequently suffer from **NEGATIVE TRANSFER**: the unconscious regression to motor habits from the previously flown aircraft type.

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Algoritmo Paso a Paso (Guía para Dummies): SHELL Model & Error Analysis

REGLA MNEMOTÉCNICA: L-H-S-E-L Paso 1: Identificar siempre el Liveware Central (L) como el piloto/operador humano. Paso 2: Si el problema involucra mandos, pantallas o botones confusos -> Es un desajuste L-H (Hardware). Paso 3: Si involucra una checklist mal redactada o un manual confuso -> Es un desajuste L-S (Software). Paso 4: Si involucra turbulencia, calor o hipoxia en cabina -> Es un desajuste L-E (Environment). Paso 5: Si involucra mala coordinación entre Capitán y Primer Oficial o con ATC -> Es un desajuste L-L (Liveware).

5. AVIATION EXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: What is the capacity and typical retention duration of human working (short-term) memory without active rehearsal?

- [A]: Unlimited capacity, lasting several hours
- [B]: 7 ± 2 chunks of information, lasting approximately 10 to 20 seconds
- [C]: Exactly 3 items, lasting 5 minutes
- [D]: 15 items, lasting 1 hour

[?] OFFICIAL RATIONALE & ANALYSIS - Official Rationale & Exam Strategy

CORRECT: [B]. Miller's Law established that short-term memory holds 7 ± 2 chunks of information with a decay time of 10 to 20 seconds without rehearsal.

Q2: In aviation psychology, what is 'chunking' in working memory?

- [A]: Forgetting unimportant flight data
- [B]: Grouping individual data items into larger, meaningful operational units to expand working memory capacity
- [C]: Memorizing checklists by rote
- [D]: Visualizing aerodynamic airflow

[?] OFFICIAL RATIONALE & ANALYSIS - Official Rationale & Exam Strategy

CORRECT: [B]. Chunking aggregates discrete pieces of information into cohesive units, effectively multiplying the storage capability of working memory.

Q3: What cognitive phenomenon describes a pilot focusing exclusively on a burnt-out landing gear bulb while

- [A]: Sensory adaptation
- [B]: Cognitive tunneling (perceptual tunneling / channelized attention)
- [C]: Episodic retrieval
- [D]: Empty field myopia

[?] OFFICIAL RATIONALE & ANALYSIS - Official Rationale & Exam Strategy

CORRECT: [B]. Cognitive tunneling occurs when high stress or fascination causes attentional narrowing onto one problem, ignoring peripheral critical cues.

Q4: Which memory system stores automated motor flying skills, such as cycling rudder pedals during a cross

- [A]: Semantic memory
- [B]: Episodic memory
- [C]: Procedural memory
- [D]: Echoic memory

[?] OFFICIAL RATIONALE & ANALYSIS - Official Rationale & Exam Strategy

CORRECT: [C]. Procedural memory stores automated physical skills and motor programs that operate largely outside conscious working memory.

Q5: What is 'Negative Habit Transfer' during aircraft type transition?

- [A]: The inability to learn new procedures
- [B]: The unconscious reversion to motor actions or procedures from a previous aircraft type under stress
- [C]: Disliking a new cockpit layout
- [D]: Forgetting radio phraseology

[?] OFFICIAL RATIONALE & ANALYSIS - Official Rationale & Exam Strategy

CORRECT: [B]. Negative transfer occurs when previously ingrained motor habits override newly acquired procedures during moments of high workload or stress.

Q6: How long does raw auditory information remain in the human echoic sensory store?

- [A]: 0.1 seconds
- [B]: 2 to 4 seconds
- [C]: 30 seconds
- [D]: Several minutes

[?] OFFICIAL RATIONALE & ANALYSIS - Official Rationale & Exam Strategy

CORRECT: [B]. Echoic (auditory) sensory store retains sounds for 2 to 4 seconds, allowing the brain to process spoken ATC clearances just heard.